

1 Frozen cohort

Use the prespecified matched cohort and frozen preprocessing outputs.

Create 5 repeats $\times$ 5 stratified outer folds.

2 Outer analysis

For each of 25 outer splits, reserve the assessment fold.

All model fitting for that split uses only its outer analysis set.

3 Inner tuning

Use three inner folds for elastic net and XGBoost.

For screened proteomics, fit limma independently in each inner analysis fold and apply it to the paired inner assessment fold.

4 Outer predictions

Choose hyperparameters; rerun limma on the full outer analysis set; refit and predict its assessment fold.

Fit LR age-only on that same outer analysis set and predict probabilities.

5 Repeated OOF

Each participant receives one out-of-fold probability per repeat.

Average the five probabilities per participant for each model.

6 Nested evaluation of stacked models

Within each repeat, combine cross-fitted clinical and omics probabilities.

Tune ridge-logistic stacks using nested resampling and retain out-of-fold ensemble probabilities; no participant predicts itself.

7 Secondary discrimination summary

Calculate ROC/AUC from averaged repeated OOF probabilities with death as the fixed positive event.

Paired DeLong comparisons use predictions on the same participants but are reported as descriptive because repeats induce dependence and this is a secondary analysis.